

## **Experiment 18:**

### **Implement Perceptron based IRIS classification**

#### **Python Code:**

```
from sklearn.datasets import load_iris
from sklearn.linear_model import Perceptron
iris = load_iris()
X, y = iris.data, iris.target
model = Perceptron(random_state=42)
model.fit(X, y)
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)[0]
print("Sample:", sample)
print("Predicted Iris Flower:", iris.target_names[prediction])
```

#### **Output:**

```
Sample: [[5.1, 3.5, 1.4, 0.2]]
Predicted Iris Flower: setosa
```

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